

Space Charge in Polypropylene Containing Synthetic NanoParticles

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Abstract- Polypropylene is extensively used as a dielectric in power capacitors, electrical machines and transformers. It has excellent mechanical, thermal, and electrical properties and provides outstanding resistance to moisture, grease, and oils. It does not present stress-cracking problems and at high temperatures it offers better thermal and chemical resistance than polyethylene. A thermoplastic such as polypropylene can be reinforced with small quantities of nanoparticles to not only achieve better chemical and thermal properties but also improved dielectric performance, such as high electrical breakdown strength, low surface and volume conductivity and resistance to partial discharges. The space charge properties of a structural nanocomposite based on layered synthetic silicate embedded in polypropylene (PP) are described in this paper. Electroluminescence emission, dissipation current measurement and pulsed electro-acoustic technique are employed to compare the properties of PP with and without the nanoparticles.

techniques cannot usually break up these agglomerates. Another problem is the immiscibility of hydrophilic nanoparticles with a hydrophobic polymer, such as PP, which results in poor interfacial interactions. Also, nanoparticles, such as natural clay, have many impurities which cannot be easily removed.

In this paper nanoparticles of synthetic clay in various concentrations are embedded into polypropylene. Electroluminescence (EL) emission, Dissipation Current Measurement (DCM) and Pulsed Electro-acoustic (PEA) technique are employed to compare the properties of PP with and without the nanoparticles. It is shown that during aging less charge is injected into the nanocomposite as compared to the conventional polypropylene. Consequently, less amount of light due to electroluminescence is emitted from the nanocomposite than the polymer without the nanoparticles.

I. INTRODUCTION

The use of nanoparticles in the matrix of polymeric materials can greatly improve the thermal, mechanical and electrical properties of polymeric composites but the area of nanodielectrics is relatively unexplored. Nanodielectrics are a class of insulating materials containing at least one phase at the nanometer scale and their benefits come in the form of increased endurance and enhanced breakdown strength of the materials [1]. It is only in the past few years that attention has been given to the application of nanocomposites in the field of electrical insulating materials [2, 3]. Nanodielectrics can last longer than conventional composites and thus extend the life of high voltage equipment and reduce the need for maintenance. Due to their superior breakdown strength and thermal properties nanodielectrics can be used more efficiently at higher fields and temperatures and this would greatly decrease equipment cost and save energy.

Nanofillers have been reported to mitigate more efficiently the space charge formation in polymeric materials [4]. The dielectric performance of nanodielectrics is primarily controlled by the interface (which is a nanophase) between the nanoparticles and the polymer matrix. In particular, it was shown that the enhanced area of interphase changes the bonding state, which affects the interfacial properties.

The electrical properties of nanodielectrics may reach their optimum value when the nanoparticles are well dispersed. One of the problems is that the nanoparticles agglomerate easily because of the high surface energy and conventional mixing

II. EXPERIMENTAL

A. Sample Preparation

Three materials, isotactic PP with 0, 2, and 4-wt% of organosilicate, referred to in this text as PP-0%, PP-2%, and PP-4%, respectively, were tested. The manufacture of polypropylene containing nanoparticles involved three steps:

1. Preparation of the Master Batch (MB)
2. Dilution of MB
3. Preparation of films

Step 1. Preparation of the Master Batch (MB).

PP (ProFax PP HL-451H from Basell) was melt compounded with a 1:1 mixture of two compatibilizers [PP grafted with maleic anhydride (PP-MA) from Eastman (Epolene- 3015) and from Crompton (Polybond 3150)], antioxidant Irganox B-225 (from Ciba) and organoclay (Topy 4CD-Ts, from Topy Co., Ltd, i.e., synthetic tetrasilicic mica pre-intercalated with di-stearyl-di-methyl ammonium chloride). The compounding at 200°C was carried out under a blanket of dry nitrogen in a twin-screw extruder (TSE; Leistritz 34 mm, L/D = 40; High shear intensity screw, Speed: 200 rpm, T = 200°C; Throughput: 5 kg/h). The resulting MB composition was: 75.8-wt% PP + 0.2-wt% antioxidant + 16- wt% compatibilizers + 8-wt% 4CD-Ts. The organoclay dispersion was characterized by X-ray diffraction (XRD) and high-resolution transmission electron microscopy (HRTEM). The interlayer spacing was $d_{001} = 3.25$ nm; the number of clay platelets in residual stacks $N = 3.25$, and the degree of exfoliation $EX = 79\%$.

Step 2. Dilution of MB.

Samples containing 0, 1, 2 and 4-wt% organoclay were prepared. To ascertain the same compounding history of the samples all samples were extruded at 180°C under a blanket of dry nitrogen, using the same TSE with the same processing parameters as for MB. Thus, PP + antioxidant were extruded twice; the MB obtained in Step 1 was re-extruded, and the other compositions were prepared by first dry-blending PP with MB at appropriate ratio. The clay dispersion parameters were: $d_{001} = 3.45$ nm, $N = 2.64$, and $EX = 88\%$, i.e., 20% above that of MB.

Step 3. Preparation of films.

The pelletized dried compositions obtained in step 2 were film blown at $T = 180^\circ\text{C}$ to 230°C into films with 135 μm thickness. Subsequently, the films were rolled twice at 115°C between 3-rolls in a calender, to obtain the final samples for testing.

B. Measurements

Samples having a diameter of 30 mm were cut from the films and each side of the sample was sputtered with gold electrodes having a diameter and thickness of 25 mm and 30 nm, respectively.

EL and DCM techniques were used to measure the electric field at which charge injection occurs in the material. Space charge distribution was measured with the PEA technique before and after aging the samples. The EL, DCM and PEA techniques have been described elsewhere [5].

III. RESULTS AND DISCUSSION

Typical number of EL pulses emitted from the three materials subjected to ac electric field up to 24 kV/mm is shown in Fig. 1. EL emission is due to the recombination of electrons and holes injected during each cycle of the ac voltage. Both nanomaterials, PP-2% and PP-4%, emitted less light than PP-0%. This suggests that less charge is injected into PP containing the nanoparticles than the conventional material.

To substantiate that EL emission is caused by the injection of charges into the polymer the dissipation current was also measured [5]. Typical waveforms of the dissipation current are shown in Fig. 2. At low fields the current waveforms are sinusoidal (see inset) but above ~ 14 kV/mm they become non-sinusoidal and contain harmonics. The non-sinusoidal behavior of the dissipation current is caused by charge injection into the polymeric insulation.

Since EL emission (Fig. 1) also starts at the same value of the electric field it clearly shows that EL emission is caused by the injection of charges into the polymeric insulation. However, there is no difference in the EL inception voltage of the three materials. This suggests that the nanoparticles do not affect the field at which charge injection is initiated. However, as shown below they reduce the amount of charge that is accumulated into the material during electrical aging.

The PEA technique was employed to measure the space charge distribution in the three materials. For ac voltage very long aging times are required to accumulate sufficient charge in the

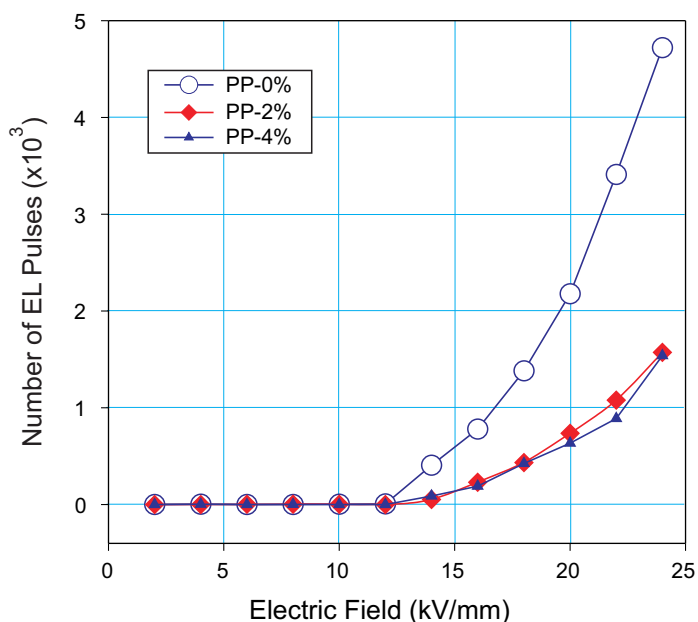


Fig. 1. EL emission in the three materials as a function of the of the ac electric field.

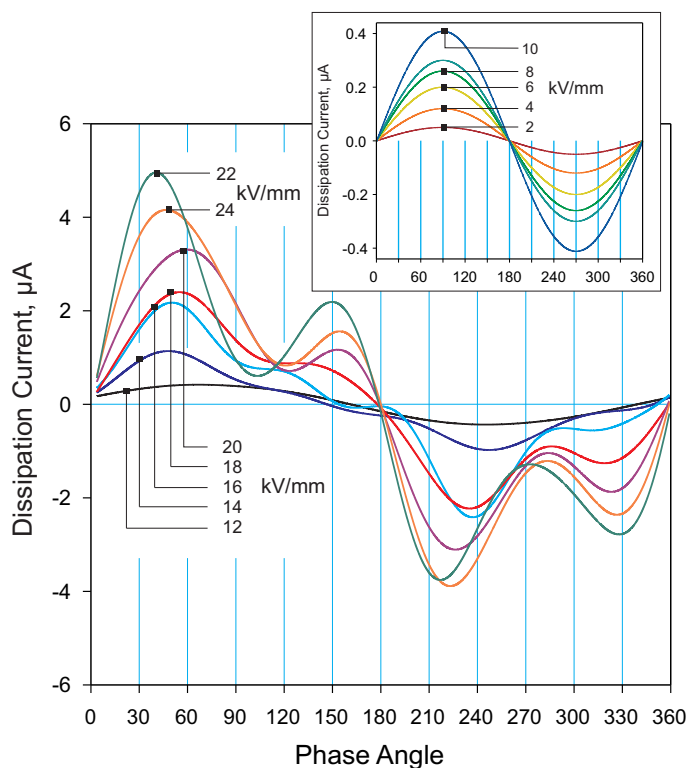


Fig. 2. Dissipation current waveforms for increasing ac electric field.

materials. This is because most of the charges injected during each half-cycle of the ac waveform are lost by recombination or extraction during the next subsequent half-cycle. Hence all three materials were aged under negative dc voltage at 40kV/mm. The space charge distribution in unaged and aged samples of PP-0%, PP-2% and PP-4% is shown in Fig. 3.

To ensure that no charge is injected during PEA measurements

the fast nanosecond pulses were applied while the unaged samples were held at a very low dc field of 0.1 kV/mm. The dotted curves show that initially there is no charge in the three unaged materials. Each sample was then aged inside the PEA test-cell at -40 kV/mm for 144 h and the space charge distribution was again monitored at the low dc field of 0.1 kV/mm. Fig. 3 shows that during aging all three materials developed a homocharge peak $\sim 15 \mu\text{m}$ from the bottom electrode and a hetrocharge peak $\sim 18 \mu\text{m}$ from the negative electrode. The polarity and amount of charge that develops during aging depends on the type of interface produced between the electrodes and the dielectric material. However, the amount of charge in PP-0% was much larger than in PP-2% and PP-4%.

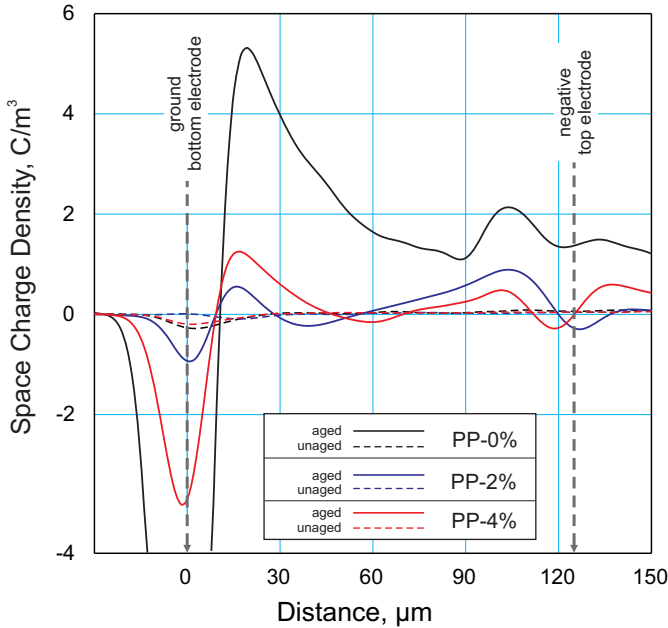


Fig. 3. Effect of 144 h of aging in dc field of -40 kV/mm, RT, on space charge distribution in the three materials.

This is the reason why the EL intensity is lower in the nanodielectrics than in the conventional material (PP-0%). Although charge injection in all three materials initiated at ~ 14 kV/mm the amount of charge injected into the nanodielectrics was smaller. Also, during aging PP-4% acquired more charge than PP-2% and this could be attributed to the uneven dispersion of the nanoparticles in the polymer. Uniform dispersion of the nanoparticles becomes difficult as their concentration increases.

During aging the nanodielectrics reduce the amount of charge that can be injected into the material. To substantiate this statement, the materials were further aged up to 500 h. Fig. 4 shows the space charge distribution in the three materials after aging for 500 h with dc field at -40 kV/mm. Again there is more charge in the conventional material than in the nanodielectrics. Comparison of Figs. 3 and 4 shows that although the amount of charge in PP-0% increases with the time of aging there is hardly any increase in the samples of PP-2% and PP-4%. For PP-2% homocharge accumulated closer to the bottom electrode and there was a hetrocharge peak $\sim 30 \mu\text{m}$ from the bottom electrode.

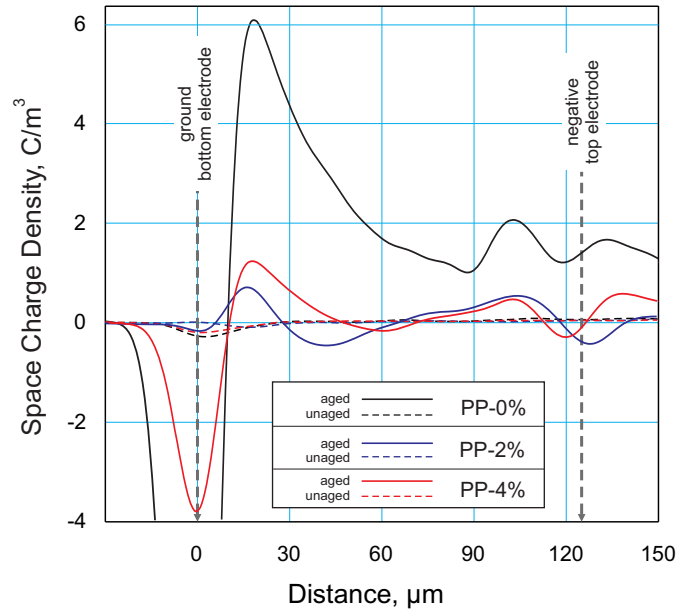


Fig. 4. Effect of 500 h of aging in dc field of -40 kV/mm, RT, on space charge distribution in the three materials.

The relatively small increase in the amount of charge in the nanodielectrics is probably due to the interfaces between the nanoparticles and the polymer. The nanoparticles have a high aspect ratio and form large interfaces with the polymer. The charges injected into the material could be trapped at these interfaces and their mobility will be decreased. This in turn will prevent any further injection of charge into the material.

IV. CONCLUSIONS

Charge injection in polypropylene with and without the nanoparticles is initiated at the same value of the electric field. The electroluminescence and dissipation current measurements showed that although charge injection was initiated at the approximately the same field in all three materials, the amount of charge injected into PP-2% and PP-4% was less than in PP-0%.

Initially there was no charge in the unaged materials but after aging for 144 h the nanodielectrics acquired less charge than the conventional materials. Even after aging the materials for 500 h, the PEA technique showed that there was only a slight increase in the amount of charge accumulated in the nanodielectrics as compared to the polypropylene without the nanoparticles. Hence, nanoparticles of organosilicates embedded into polypropylene restrict the amount of charge that can accumulate in the material. This can be very useful for power apparatus such as high voltage capacitors and transformers which operate at high electric field.

ACKNOWLEDGMENTS

The authors are indebted to Dr. L. Utracki of the Industrial Materials Institute of NRC Canada for providing nanocomposite specimens for this work and to Mrs. Y. Chen of the Institute for National Measurement Standards of NRC Canada for her invaluable technical support.

REFERENCES

- [1] Y. Cao, P. Irwin and K. Younsi, "The Future of Nanodielectrics in the Electrical Power Industry" IEEE Trans. Dielectr. Electr. Insul., Vol. 11, pp. 797-807, 2004.
- [2] Y. Cao, and P.C. Irwin, "The Electrical Conduction in Polyimide Nanocomposites" IEEE CEIDP, Albuquerque, USA, pp. 116-119, 2003.
- [3] H. Z. Ding and R. Varlow, "Effect of Nano-Fillers on Electrical Treeing in Epoxy Resin Subjected to AC Voltage", IEEE CEIDP, pp. 332-335, 2004.
- [4] Y. Yin, J. Chen, J. Yang, D. Xiao, D. Tu, R. Yin and H. Qian, "Effect of Space Charge in Nanocomposites of LDPE/TiO₂" IEEE 7th Intern. Conference on Properties and Application of Dielectric Materials, Nagoya, pp. 913-916, 2003.
- [5] S. Bamji, M. Abou-Dakka, A. Bulinski and L. Utracki, "Electroluminescence and Space Charge in Nanodielectrics Subjected to AC Voltage", The Journal of Central Power Research Institute, Bengaluru, India, Vol. 4, No. 2, pp. 185-194, Sep. 2008.